

Session 1 · Deck 1.A3

AI: Tool *and* Object of Macro Research

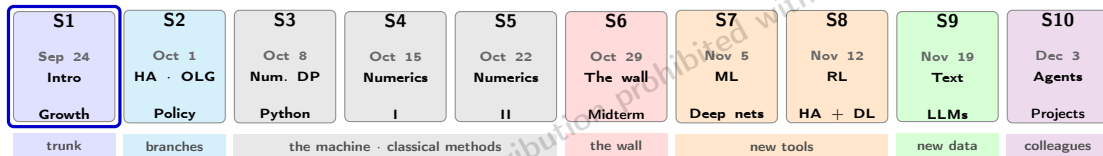
Zhigang Feng

Thursday, September 24, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

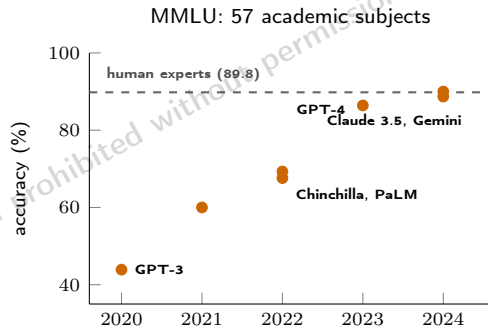
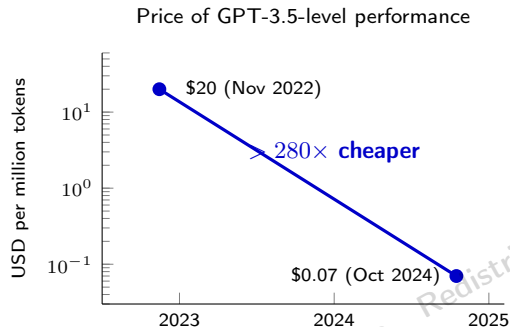
▼ you are here: 1.A3



Build it on paper → solve it on the machine → push it to the frontier with AI

This deck: AI as an *object* the macroeconomy must absorb, AI as a *tool* for macro research, and how this course turns both into your practice.

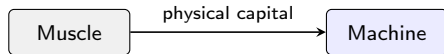
A Number to Start: Cognition Got $280\times$ Cheaper in Two Years



Stanford AI Index 2025; reported model scores (MMLU, Hendrycks et al. 2021). Benchmarks saturate quickly — treat any snapshot as a moving target.

The economist's question: when a general-purpose input becomes this cheap this fast, what happens to *tasks*, *wages*, the *tax base* — and to the models we write?

AI as an Object: A New Kind of Capital



Industrial revolution: substitute for *physical* effort



Now: substitute for (and complement to) *cognitive* effort

What makes it unusual capital?

- *Non-rival*: one model serves millions at once.
- *Replicable*: copying weights takes seconds; training a worker takes decades.
- *Intangible*: weights, data, pipelines — poorly measured in national accounts.
- *Self-improving*: it generates code and data that improve AI.

A general-purpose technology? Pervasive, improving, spawning complements (Bresnahan & Trajtenberg 1995) — but past GPTs paid off only with long lags (David 1990).

And it acts. Copies of one agent *coordinate*: one objective, machine-speed communication. And that objective only approximates yours — **alignment is a principal-agent problem**, not a technology parameter.

Where AI Enters Production: Tasks and the Adoption Margin

Task-level production. In task i , human labor h_i and AI services m_i combine with elasticity σ_i :

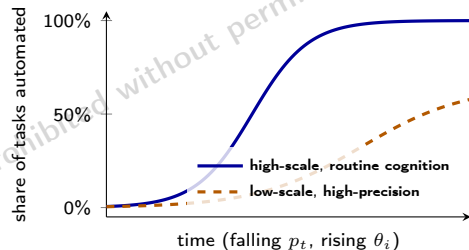
$$H_i = \left[\varphi_i h_i^{\frac{\sigma_i-1}{\sigma_i}} + (1 - \varphi_i) (\theta_i m_i)^{\frac{\sigma_i-1}{\sigma_i}} \right]^{\frac{\sigma_i}{\sigma_i-1}}$$

θ_i : AI capability in task i ; σ_i : how substitutable AI is for people.

The adoption margin. Automate task i when the value exceeds the up-front cost:

$$\frac{(w_i - p_t) \cdot \text{Scale}_i}{\rho} \geq C_i^{\text{upfront}}$$

with wage w_i , AI price p_t , discount rate ρ ; cost rises steeply with required precision.



- Tasks cross the threshold *in order*: an S-curve with a long tail.

Does It Add Up? From Tasks to Macro

- Growth accounting:
 $\Delta \ln Y = \alpha \Delta \ln K + (1 - \alpha) \Delta \ln L + \Delta \ln A$. AI can show up in K (automation), in A (ideas, faster R&D), or as an *unmeasured* intangible stock.
- Task gains aggregate with weights — and bottlenecks:

$$\Delta A_{\text{macro}} \approx \sum_i \omega_i \Delta a_i.$$

If heavily weighted tasks do not budge, automating the easy ones barely moves TFP (Baumol / weakest link).

A sober benchmark

Acemoglu (2024), *The Simple Macroeconomics of AI*: a TFP gain of no more than about **0.66% over ten years**, because only about 4.6% of tasks are likely to be affected in that window.

The research opening

Whether AI is a slow J-curve or a growth break is an **open quantitative question** — and so is *who* gains. Answering it needs dynamic GE models with heterogeneity.

AI as an Object: The Questions Macroeconomists Must Answer

Labor and inequality

Which tasks, which workers? Reallocation across tasks, polarization, returns to skill — e.g. $\Lambda(z) = [\sigma_T - \sigma(z)]s_K(z)$ from 1.A1.

Public finance

If income shifts from labor to AI capital, what happens to the tax base and to social insurance? Should compute be taxed, and how?

Growth and measurement

Non-rival intangible capital, diffusion lags, mismeasured output; is AI a level shift or a growth-rate shift?

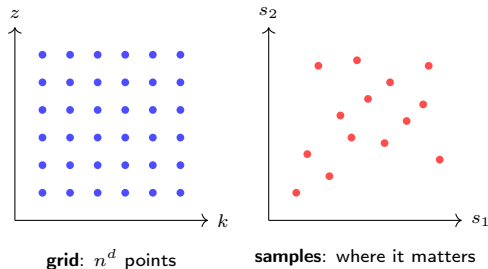
Money, finance, stability

Algorithmic pricing and faster information: inflation dynamics, expectations, financial fragility.

Research in Practice

Each of these is a **dynamic general-equilibrium question with heterogeneity** — exactly the models of 1.A2. That is why the economics of AI is becoming central to the *macro* research frontier.

AI as a Tool (1): Solving Models We Could Not Solve



- Replace grids by **samples** and polynomials by **neural networks** v_θ, g_θ .
- Train on the model's own **equilibrium conditions**: Euler and Bellman residuals, market clearing.
- Use **reinforcement learning** when the state is huge — e.g. a whole distribution.

Research in Practice

The frontier, briefly: deep learning for dynamic models (Maliar, Maliar & Winant 2021); deep equilibrium nets (Azinovic, Gaegauf & Scheidegger 2022); DeepHAM (Han, Yang & E); DeepSAM for search and matching (Payne, Rebei & Yang 2024); optimal taxation without commitment with a distribution as state (Feng, Han, Sargent & Zhu 2026). *We build up to all of these in S7–S8.*

AI as a Tool (2): A Research Collaborator

Case: extending a deep Ramsey-taxation solver from a representative agent to heterogeneous agents (Chien, Feng & Wen, in progress). The theory → code → debug loop shrank from months to days.

Contribution	AI	Economist
Drafting LaTeX, vectorized PyTorch	✓	
Proposing penalty and boundary-learning designs	✓	
Summarizing diagnostics, code–paper mis-matches	✓	
Checking economic correctness of the model		✓
Catching timing and transition-direction bugs		✓
Choosing benchmarks; accepting results		✓

AI supplies throughput; the economist supplies judgment.

The failures were the most valuable part: each forced a better idea — and each was caught by a person who knew what the answer should look like.

AI as a Tool (3): New Data, New Workflows — and One Guardrail

Where AI excels

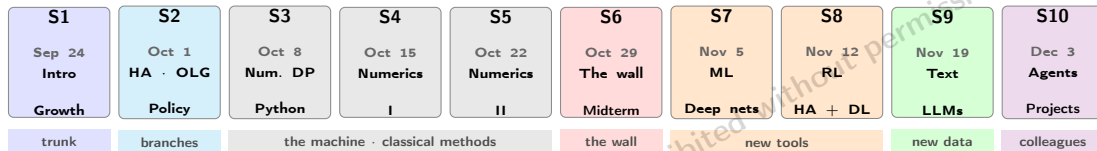
- turning text into data: FOMC statements, filings, news (S9);
- approximating high-dimensional functions (S7–S8);
- writing, refactoring, and debugging code at scale (S10);
- searching and summarizing a literature.

Where economists add value

- the question, and the right state variables;
- identification: *why* a number is causal;
- welfare and normative judgment;
- deciding whether an output can be *defended*.

The guardrail. Prediction $\neq$ causation, and a plausible answer $\neq$ a correct one. As generating answers gets cheap, **verification becomes the scarce skill** — the one this course trains hardest.

The Ten-Week Plan



Build it on paper → solve it on the machine → push it to the frontier with AI

Running thread	Path through the course
The Aiyagari economy	theory (S2) → classical solution (S6) → deep-learning solution (S8) → an agent audits a solver (S10)
Taxation without commitment	Ramsey vs. Markov-perfect (S2) → why it is hard (S6) → neural solution (S8)
AI and inequality	AI as capital (S1) → measuring exposure with language models (S9) → the future of research (S10)

You Are the Research Architect

The 5-pillar workflow, on every problem:

1. **Economic theory** — objective, constraints, equilibrium.
2. **Algorithm design** — which method, and why.
3. **Numerical technique** — grids, quadrature, sampling.
4. **Trade-offs** — accuracy vs. speed, bias vs. variance.
5. **Code literacy & validation** — read every line; benchmark; sanity-check the economics.

Replicate first, explore second

For every headline number:

1. re-derive one number by hand or in a second tool;
2. compare to a *known* benchmark;
3. run economic sanity checks (signs, Euler errors, market clearing);
4. never submit code you have not read end to end.

AI can lay the bricks. **You own the design, the validation, and the interpretation** — and you must be able to defend all three.

How the Course Works

Component	Weight	Dates
HW 1: dynamic programming & numerical methods	15%	out Oct 8 / 15; due Oct 22
HW 2: heterogeneity & learning	15%	out Oct 29; due Nov 12
Midterm (closed book, in class)	20%	Oct 29, Block B
Final project (teams of 2–3)	40%	proposal Nov 16; talk Dec 3; paper Dec 10
Participation	10%	every session

Each session: two 75-minute blocks, three short decks each, a live demo, and an exit ticket.

AI policy

- **Homework and project:** AI tools are encouraged. Keep an `AI_LOG.md`: the prompts that mattered, where the AI was wrong, what you verified.
- You must be able to explain every line you submit.
- **Midterm:** closed book, no devices.

Final Project: A Mini-Conference on December 3

Track	Replicate a benchmark, then extend it
A	A quantitative model (Aiyagari, Krusell-Smith, OLG) and a policy experiment
B	A classically hard model solved with deep learning or RL, validated against a benchmark
C	Economics of AI: a task-based model, or an exposure measure linked to outcomes
D	Text or LLM measurement that feeds a macro model or its calibration

Milestones

- **Nov 5:** teams and topic
- **Nov 16:** two-page proposal
- **Dec 3:** presentation, plus discussing another team's paper
- **Dec 10:** short paper (≤ 10 pages) and replication code

Start looking now: every deck ends with a thread you could pull into a project.

Bridge: Back to the Trunk

- AI is an **object** — non-rival cognitive capital whose effects on tasks, wages and the tax base are open macro questions.
- AI is a **tool** — for solving models past the computational wall, measuring the economy, and running research — as long as *you* verify.
- The course runs from paper, to machine, to frontier; the grade rewards **correct, defended** results.

After the break (Block B): the optimal growth model.

- From Solow to the planner; Euler equations and steady states.
- The recursive formulation and why dynamic programming works.
- A platform, not a model: how every branch of 1.A2 grows from it.

Exit ticket (end of today): one thing that clicked, one thing that did not. It takes one minute and it shapes next week.